

# SOLIDWORKS API ( .NET VBA C# ) — Reference (204/337): IModeler (part 2 of 4)

SOLIDWORKS API Documentation (converted from CHM)

## IModeler (part 2 of 4) (Type)

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### CreatePCurve Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreatePCurve Method (IModeler)

Creates a pcurve.

#### .NET Syntax

##### Visual Basic (Declaration)

```
Function CreatePCurve( _  
    ByVal Surface As System.Object, _  
    ByVal Props As System.Object, _  
    ByVal Knots As System.Object, _  
    ByVal CtrlPtCoords As System.Object _  
) As System.Object
```

##### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Surface As System.Object  
Dim Props As System.Object  
Dim Knots As System.Object  
Dim CtrlPtCoords As System.Object  
Dim value As System.Object  
  
value = instance.CreatePCurve(Surface, Props, Knots, CtrlPtCoords)
```

##### C#

```
System.object CreatePCurve(  
    System.object Surface,  
    System.object Props,  
    System.object Knots,  
    System.object CtrlPtCoords  
)
```

##### C++/CLI

```
System.Object^ CreatePCurve(  
& System.Object^ Surface,  
& System.Object^ Props,  
& System.Object^ Knots,  
& System.Object^ CtrlPtCoords  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

[Surface](#)  
[Surface](#) associated with the pcurve  
[Props](#)  
Array containing 4 integers packed into 2 double elements (see **Remarks**)  
[Knots](#)  
Array of numKnots (see **Remarks**)  
[CtrlPtCoords](#)  
Array of numCtrlPtCoord (see **Remarks**)

#### Return Value

[Pcurve](#)

#### Visual Basic for Applications (VBA) Syntax

See [Modeler::CreatePCurve](#).

#### Example

## Remarks

The Props argument contains the following values:

- DimensionControlPoints
- Order
- NumCtrlPoints
- Periodicity

| Length of this array... | Given by...                                             |
|-------------------------|---------------------------------------------------------|
| Knots                   | numKnots = NumCtrlPoints + Order                        |
| CtrlPtCoords            | NumCtrlPtCoord = NumCtrlPoints * DimensionControlPoints |

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::ICreatePCurve Method](#)  
[ICurve::GetPCurveParams Method](#)  
[ICurve::IGetPCurveParams Method](#)  
[ICurve::IGetPCurveParamsSize Method](#)  
[IModeler::CreateBsplineCurve Method](#)  
[IModeler::ICreateBsplineCurve Method](#)

## Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

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## CreatePlanarSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreatePlanarSurface Method (IModeler)

Obsolete. Superseded by [IModeler::CreatePlanarSurface2](#).

## .NET Syntax

Visual Basic (Declaration)

```
Function CreatePlanarSurface( _  
    ByVal VRootPoint As System.Object, _  
    ByVal VNormal As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim VRootPoint As System.Object  
Dim VNormal As System.Object  
Dim value As System.Object
```

```
value = instance.CreatePlanarSurface(VRootPoint, VNormal)
```

C#

```
System.object CreatePlanarSurface(  
    System.object VRootPoint,  
    System.object VNormal  
)
```

C++/CLI

```
System.Object^ CreatePlanarSurface(  
& System.Object^ VRootPoint,
```

```
& System.Object^ VNormal
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*VRootPoint*

*VNormal*

### Visual Basic for Applications (VBA) Syntax

See [Modeler::CreatePlanarSurface](#).

### See Also

[IModeler Interface](#)  
[IModeler Members](#)

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## CreatePlanarSurface2 Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreatePlanarSurface2 Method (IModeler)

Creates a new infinite planar surface.

### .NET Syntax

#### Visual Basic (Declaration)

```
Function CreatePlanarSurface2(
    ByVal VRootPoint As System.Object, _
    ByVal VNormal As System.Object, _
    ByVal VRef As System.Object _
) As System.Object
```

#### Visual Basic (Usage)



```
Dim instance As IModeler
Dim VRootPoint As System.Object
Dim VNormal As System.Object
Dim VRef As System.Object
Dim value As System.Object
```

```
value = instance.CreatePlanarSurface2(VRootPoint, VNormal, VRef)
```

#### C#

```
System.object CreatePlanarSurface2(
    System.object VRootPoint,
    System.object VNormal,
    System.object VRef
)
```

#### C++/CLI

```
System.Object^ CreatePlanarSurface2(
    & System.Object^ VRootPoint,
    & System.Object^ VNormal,
    & System.Object^ VRef
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*VRootPoint*

Array of 3 doubles (x,y,z) representing the root point of the VNormal and VRef vectors

*VNormal*

Array of 3 doubles (x,y,z) defining a vector with root point, VRootPoint; this vector must be perpendicular to VRef

*VRef*

Array of 3 doubles (x,y,z) defining the x-axis on the planar surface with root point, VRootPoint; this vector must be perpendicular to VNormal

#### Return Value

Newly created planar [surface](#)

### Visual Basic for Applications (VBA) Syntax

See [Modeler::CreatePlanarSurface2](#).

### Example

[Create Infinite Plane \(VBA\)](#)  
[Cut Body in Half Using Macro Feature \(VBA\)](#)  
[Create Temporary Extruded Body \(C#\)](#)  
[Create Temporary Extruded Body \(VB.NET\)](#)  
[Create Temporary Extruded Body \(VBA\)](#)

### Remarks

You can use this method with:

- A set of related functions designed to construct a body from trimmed surfaces. It can also be used with trimming curve creation routines to construct a trimmed planar surface.
- Trimming curve creation routines such as [ISurface::AddTrimmingLoop2](#) or [ISurface::IAddTrimmingLoop2](#) to construct a trimmed surface

See [IBody2::CreatePlanarTrimSurfaceDLL](#) or [IBody2::ICreatePlanarTrimSurfaceDLL](#), which allows you to create a planar trimmed surface directly from the vertex points.

### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateBsplinesSurface Method](#)  
[IModeler::CreateConicalSurface2 Method](#)  
[IModeler::CreateCoonsBSurface Method](#)  
[IModeler::CreateCylindricalSurface2 Method](#)  
[IModeler::CreateExtrusionSurface Method](#)  
[IModeler::CreateLoftSurface Method](#)  
[IModeler::CreateOffsetSurface Method](#)  
[IModeler::CreateRuledSurface Method](#)  
[IModeler::CreateSphericalSurface2 Method](#)  
[IModeler::CreateSweptSurface Method](#)  
[IModeler::CreateToroidalSurface Method](#)  
[IModeler::ICreateBsplinesSurface Method](#)  
[IModeler::ICreateConicalSurface2 Method](#)  
[IModeler::ICreateCylindricalSurface2 Method](#)  
[IModeler::ICreateExtrusionSurface Method](#)  
[IModeler::ICreateLoftSurface Method](#)  
[IModeler::ICreateOffsetSurface Method](#)  
[IModeler::ICreatePlanarSurface2 Method](#)  
[IModeler::ICreateRuledSurface Method](#)  
[IModeler::ICreateSphericalSurface2 Method](#)  
[IModeler::ICreateSweptSurface Method](#)  
[IModeler::ICreateToroidalSurface Method](#)

### Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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## CreateRuledSurface Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateRuledSurface Method (IModeler)

Creates a ruled surface from the curves.

### .NET Syntax

Visual Basic (Declaration)

```
Function CreateRuledSurface( _  
    ByVal Curve1In As System.Object, _  
    ByVal Curve2In As System.Object, _  
    ByVal Apex As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```

Dim instance As IModeler
Dim Curve1In As System.Object
Dim Curve2In As System.Object
Dim Apex As System.Object
Dim value As System.Object

value = instance.CreateRuledSurface(Curve1In, Curve2In, Apex)

```

C#

```

System.object CreateRuledSurface(
    System.object Curve1In,
    System.object Curve2In,
    System.object Apex
)

```

C++/CLI

```

System.Object^ CreateRuledSurface(
    & System.Object^ Curve1In,
    & System.Object^ Curve2In,
    & System.Object^ Apex
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

[Curve1In](#)  
 First [curve](#)  
[Curve2In](#)  
 Second [curve](#)  
[Apex](#)  
 Array of 3 doubles, (x, y, z), the apex point

#### Return Value

Newly created ruled [surface](#)

### Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateRuledSurface](#).

### Example

[Create Ruled Surface \(VBA\)](#).

### Remarks

Valid input curves are b-curves.

### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateBsplinesSurface Method](#)  
[IModeler::CreateConicalSurface2 Method](#)  
[IModeler::CreateCoonsBSurface Method](#)  
[IModeler::CreateCylindricalSurface2 Method](#)  
[IModeler::CreateExtrudedBody Method](#)  
[IModeler::CreateExtrusionSurface Method](#)  
[IModeler::CreateLoftSurface Method](#)  
[IModeler::CreateOffsetSurface Method](#)  
[IModeler::CreatePlanarSurface2 Method](#)  
[IModeler::CreateSphericalSurface2 Method](#)  
[IModeler::CreateSweptSurface Method](#)  
[IModeler::CreateToroidalSurface Method](#)  
[IModeler::ICreateBsplinesSurface Method](#)  
[IModeler::ICreateConicalSurface2 Method](#)  
[IModeler::ICreateCylindricalSurface2 Method](#)  
[IModeler::ICreateExtrusionSurface Method](#)  
[IModeler::ICreateLoftSurface Method](#)  
[IModeler::ICreateOffsetSurface Method](#)  
[IModeler::ICreatePlanarSurface2 Method](#)  
[IModeler::ICreateRuledSurface Method](#)  
[IModeler::ICreateSheetFromSurface2 Method](#)  
[IModeler::ICreateSphericalSurface2 Method](#)  
[IModeler::ICreateSweptSurface Method](#)  
[IModeler::ICreateToroidalSurface Method](#)

[IModeler::CreateRuledSurfaceFromEdge Method](#)  
[IModeler::ICreateRuledSurfaceFromEdge Method](#)  
[IFeatureManager::InsertRuledSurfaceFromEdge2 Method](#)

## Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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## CreateRuledSurfaceFromEdge Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateRuledSurfaceFromEdge Method (IModeler)

Creates a ruled surface using the specified edges and returns a surface body.

## .NET Syntax

Visual Basic (Declaration)

```
Function CreateRuledSurfaceFromEdge( _  
    ByVal ModDoc As ModelDoc2, _  
    ByVal Edges As System.Object, _  
    ByVal Type As System.Integer, _  
    ByVal Length As System.Double, _  
    ByVal FlipPullDir As System.Boolean, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal TrimAndSew As System.Boolean, _  
    ByVal Angle As System.Double, _  
    ByVal CoordInput As System.Boolean, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double, _  
    ByVal BConnectSurface As System.Boolean, _  
    ByRef RuledSurface As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim ModDoc As ModelDoc2  
Dim Edges As System.Object  
Dim Type As System.Integer  
Dim Length As System.Double  
Dim FlipPullDir As System.Boolean  
Dim FlipDir As System.Boolean  
Dim TrimAndSew As System.Boolean  
Dim Angle As System.Double  
Dim CoordInput As System.Boolean  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim BConnectSurface As System.Boolean  
Dim RuledSurface As System.Object  
Dim value As System.Integer
```

```
value = instance.CreateRuledSurfaceFromEdge(ModDoc, Edges, Type, Length, FlipPullDir, FlipDir,  
TrimAndSew, Angle, CoordInput, X, Y, Z, BConnectSurface, RuledSurface)
```

C#

```
System.int CreateRuledSurfaceFromEdge(  
    ModelDoc2 ModDoc,  
    System.object Edges,  
    System.int Type,  
    System.double Length,  
    System.bool FlipPullDir,  
    System.bool FlipDir,  
    System.bool TrimAndSew,  
    System.double Angle,
```

```

        System.bool CoordInput,
        System.double X,
        System.double Y,
        System.double Z,
        System.bool BConnectSurface,
        out System.object RuledSurface
    )

```

C++/CLI

```

System.int CreateRuledSurfaceFromEdge(
    & ModelDoc2^ ModDoc,
    & System.Object^ Edges,
    & System.int Type,
    & System.double Length,
    & System.bool FlipPullDir,
    & System.bool FlipDir,
    & System.bool TrimAndSew,
    & System.double Angle,
    & System.bool CoordInput,
    & System.double X,
    & System.double Y,
    & System.double Z,
    & System.bool BConnectSurface,
    & [Out] System.Object^ RuledSurface
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*ModDoc*

[Model document](#) in which to create surface body

*Edges*

Array of edges on which to create ruled surface

**NOTE:** Currently only a single edge is supported.

*Type*

- 0 = Tangent to Surface
- 1 = Normal to Surface
- 2 = Tapered to Vector
- 3 = Perpendicular to Vector
- 4 = Sweep

*Length*

Distance at which to create the surface; valid for Tangent to Surface, Tapered to Vector, Perpendicular to Vector, and Sweep types only

*FlipPullDir*

True to flip the pull direction, false to not; valid for Normal to Surface and Tapered to Vector types only

*FlipDir*

True to flip the direction, false to not; valid for Perpendicular to Vector type only

*TrimAndSew*

True to trim and knit the surface, false to not

*Angle*

Angle for Tapered to Vector type only

*CoordInput*

True to enable coordinate input, false if not; for Sweep type only

*X*

x coordinate

*Y*

y coordinate

*Z*

z coordinate

*BConnectSurface*

True to remove any connecting surfaces, false to not

*RuledSurface*

Ruled surface [body](#)

#### Return Value

0 if ruled surface body created, -1 if not

See [Modeler::CreateRuledSurfaceFromEdge](#).

## Example

[Create Ruled Surface Body From Edge \(VBA\)](#)

## Remarks

This method returns a surface body, unlike [IFeatureManager::InsertRuledSurfaceFromEdge2](#), which returns a feature.

If you [select](#) an edge with a selection mark of 4, the default face is used. If you select an edge with a selection mark of 6, the alternate face is used.

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::ICreateRuledSurfaceFromEdge Method](#)  
[IModeler::ICreateRuledSurface Method](#)  
[IModeler::CreateRuledSurface Method](#)

## Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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## CreateSheetFromFaces Method (IModeler)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateSheetFromFaces Method (IModeler)

Creates a sheet (surface) body from connected faces.

## .NET Syntax

### Visual Basic (Declaration)

```
Function CreateSheetFromFaces( _  
    ByVal FaceArr As System.Object _  
) As Body2
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim FaceArr As System.Object  
Dim value As Body2  
  
value = instance.CreateSheetFromFaces(FaceArr)
```

### C#

```
Body2 CreateSheetFromFaces(  
    System.object FaceArr  
)
```

### C++/CLI

```
Body2^ CreateSheetFromFaces(  
& System.Object^ FaceArr  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*FaceArr*  
Array of connected [faces](#)

### Return Value

Newly created [body](#)

## Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateSheetFromFaces](#).

## Example

[Create Sheet Body From Faces and Feature From Sheet Body \(VBA\)](#)

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateSheetFromSurface Method](#)  
[IModeler::ICreateSheetFromFaces Method](#)  
[IModeler::ICreateSheetFromSurface2 Method](#)  
[IFace2::CreateSheetBody Method](#)  
[IFace2::CreateSheetBodyByFaceExtension Method](#)  
[IFace2::ICreateSheetBody Method](#)  
[IFace2::ICreateSheetBodyByFaceExtension Method](#)

## Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

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## CreateSheetFromSurface Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateSheetFromSurface Method (IModeler)

Creates a sheet (surface) body from a surface.

## .NET Syntax

### Visual Basic (Declaration)

```
Function CreateSheetFromSurface( _  
    ByVal SurfaceIn As System.Object, _  
    ByVal UvRange As System.Object _  
) As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim SurfaceIn As System.Object  
Dim UvRange As System.Object  
Dim value As System.Object  
  
value = instance.CreateSheetFromSurface(SurfaceIn, UvRange)
```

### C#

```
System.object CreateSheetFromSurface(  
    System.object SurfaceIn,  
    System.object UvRange  
)
```

### C++/CLI

```
System.Object^ CreateSheetFromSurface(  
& System.Object^ SurfaceIn,  
& System.Object^ UvRange  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*SurfaceIn*

[Surface](#) from which you want to create this sheet body

*UvRange*

Array of UV values for this surface

## Return Value

Newly created [body](#).

## Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateSheetFromSurface](#).

## Example

[Cut Body in Half Using Macro Feature \(VBA\)](#)

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateSheetFromFaces Method](#)  
[IModeler::ICreateSheetFromFaces Method](#)  
[IModeler::ICreateSheetFromSurface2 Method](#)  
[ISurface::CreateTrimmedSheet Method](#)

## Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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## CreateSphericalSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateSphericalSurface Method (IModeler)

Obsolete. Superseded by [IModeler::CreateSphericalSurface2](#).

## .NET Syntax

### Visual Basic (Declaration)

```
Function CreateSphericalSurface( _  
    ByVal Center As System.Object, _  
    ByVal Radius As System.Double _  
) As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Object  
Dim Radius As System.Double  
Dim value As System.Object  
  
value = instance.CreateSphericalSurface(Center, Radius)
```

### C#

```
System.object CreateSphericalSurface(  
    System.object Center,  
    System.double Radius  
)
```

### C++/CLI

```
System.Object^ CreateSphericalSurface(  
& System.Object^ Center,  
& System.double Radius  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Parameters

*Center*

*Radius*

## Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateSphericalSurface](#).

## See Also

[IModeler Interface](#)  
[IModeler Members](#)

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## CreateSphericalSurface2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateSphericalSurface2 Method (IModeler)

Creates an untrimmed spherical surface.

## .NET Syntax

### Visual Basic (Declaration)

```
Function CreateSphericalSurface2( _  
    ByVal Center As System.Object, _  
    ByVal Axis As System.Object, _  
    ByVal RefDir As System.Object, _  
    ByVal Radius As System.Double _  
) As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Object  
Dim Axis As System.Object  
Dim RefDir As System.Object  
Dim Radius As System.Double  
Dim value As System.Object  
  
value = instance.CreateSphericalSurface2(Center, Axis, RefDir, Radius)
```

### C#

```
System.object CreateSphericalSurface2(  
    System.object Center,  
    System.object Axis,  
    System.object RefDir,  
    System.double Radius  
)
```

### C++/CLI

```
System.Object^ CreateSphericalSurface2(  
& System.Object^ Center,  
& System.Object^ Axis,  
& System.Object^ RefDir,  
& System.double Radius  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*Center*  
Array containing 3 doubles for XYZ location of the center of the spherical surface  
*Axis*  
Array containing 3 doubles  
*RefDir*  
Array containing 3 doubles  
*Radius*  
Radius at the center

### Return Value

Untrimmed spherical [surface](#)

## Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateSphericalSurface2](#).

## Remarks

The axis argument is the axis of the sphere; the refDir argument is perpendicular to the axis and through the center of the sphere and defines the origin of the UV space.

You can trim the resulting surface to generate a sheet body, for example, using [ISurface::CreateTrimmedSheet](#).

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateBsplinesurface Method](#)  
[IModeler::CreateConicalSurface2 Method](#)  
[IModeler::CreateCoonsBSurface Method](#)  
[IModeler::CreateCylindricalSurface2 Method](#)  
[IModeler::CreateExtrusionSurface Method](#)  
[IModeler::CreateOffsetSurface Method](#)  
[IModeler::CreatePlanarSurface2 Method](#)  
[IModeler::CreateRuledSurface Method](#)  
[IModeler::CreateSweptSurface Method](#)  
[IModeler::CreateToroidalSurface Method](#)  
[IModeler::ICreateBsplinesurface Method](#)  
[IModeler::ICreateConicalSurface2 Method](#)  
[IModeler::ICreateCylindricalSurface2 Method](#)  
[IModeler::ICreateExtrusionSurface Method](#)  
[IModeler::ICreateOffsetSurface Method](#)  
[IModeler::ICreatePlanarSurface2 Method](#)  
[IModeler::ICreateRuledSurface Method](#)  
[IModeler::ICreateSweptSurface Method](#)  
[IModeler::ICreateToroidalSurface Method](#)

## Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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## CreateSpring Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateSpring Method (IModeler)

Creates the geometry for a spring.

## .NET Syntax

Visual Basic (Declaration)

```
Function CreateSpring() As Spring
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim value As Spring  
  
value = instance.CreateSpring()
```

C#

```
Spring CreateSpring()
```

C++/CLI

```
Spring^ CreateSpring();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

**Return Value**

[Spring](#)

## Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateSpring](#).

## See Also

[IModeler Interface](#)  
[IModeler Members](#)

## Availability

SOLIDWORKS 2005 SP1, Revision Number 13.1

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## CreateSweptBody Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateSweptBody Method (IModeler)

Creates a swept body.

## .NET Syntax

### Visual Basic (Declaration)

```
Function CreateSweptBody( _  
    ByVal PModDoc As ModelDoc2, _  
    ByVal Propagate As System.Boolean, _  
    ByVal Alignment As System.Boolean, _  
    ByVal TwistCtrlOption As System.Short, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal BAdvancedSmoothing As System.Boolean, _  
    ByVal StartMatchingType As System.Short, _  
    ByVal EndMatchingType As System.Short, _  
    ByVal IsThinBody As System.Boolean, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ThinType As System.Short, _  
    ByVal PathAlign As System.Short, _  
    ByVal TwistAngle As System.Double, _  
    ByVal BMergeSmoothFaces As System.Boolean _  
) As Body2
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim PModDoc As ModelDoc2  
Dim Propagate As System.Boolean  
Dim Alignment As System.Boolean  
Dim TwistCtrlOption As System.Short  
Dim KeepTangency As System.Boolean  
Dim BAdvancedSmoothing As System.Boolean  
Dim StartMatchingType As System.Short  
Dim EndMatchingType As System.Short  
Dim IsThinBody As System.Boolean  
Dim Thickness1 As System.Double  
Dim Thickness2 As System.Double  
Dim ThinType As System.Short  
Dim PathAlign As System.Short  
Dim TwistAngle As System.Double  
Dim BMergeSmoothFaces As System.Boolean  
Dim value As Body2
```

```
value = instance.CreateSweptBody(PModDoc, Propagate, Alignment, TwistCtrlOption, KeepTangency,  
BAdvancedSmoothing, StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2,  
ThinType, PathAlign, TwistAngle, BMergeSmoothFaces)
```

C#

```

Body2 CreateSweptBody(
    ModelDoc2 PModDoc,
    System.bool Propagate,
    System.bool Alignment,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool BAdvancedSmoothing,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType,
    System.short PathAlign,
    System.double TwistAngle,
    System.bool BMergeSmoothFaces
)

```

C++/CLI

```

Body2^ CreateSweptBody(
    & ModelDoc2^ PModDoc,
    & System.bool Propagate,
    & System.bool Alignment,
    & System.short TwistCtrlOption,
    & System.bool KeepTangency,
    & System.bool BAdvancedSmoothing,
    & System.short StartMatchingType,
    & System.short EndMatchingType,
    & System.bool IsThinBody,
    & System.double Thickness1,
    & System.double Thickness2,
    & System.short ThinType,
    & System.short PathAlign,
    & System.double TwistAngle,
    & System.bool BMergeSmoothFaces
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*PModDoc*

[Model document](#)

*Propagate*

True propagates the swept protrusion to the next tangent edge, false does not

*Alignment*

True causes the swept protrusion to go through the end faces if the curve used for the sweep goes from one face to another or from one edge to another, false causes the swept protrusion to begin and end perpendicular to the sweep curve and it cannot break through the two end faces of the body

*TwistCtrlOption*

Twist control options as defined by [swTwistControlType\\_e](#)

*KeepTangency*

If the sweep section has tangent segments, then True to cause the corresponding surfaces in the resulting sweep to be tangent, false to not

*BAdvancedSmoothing*

If the sweep section has circular or elliptical arcs, then True to approximate the sections and smooth the surfaces, false to not

*StartMatchingType*

Tangency type as defined by [swTangencyType\\_e](#)

*EndMatchingType*

Tangency type as defined by [swTangencyType\\_e](#)

*IsThinBody*

True if this feature is a thin body, false if not

*Thickness1*

Thickness value for the first direction

*Thickness2*

Thickness value for the second direction

*ThinType*

Thin wall type as defined by [swThinWallType\\_e](#)

*PathAlign*

Align path type (see **Remarks**)

*TwistAngle*

If TwistCtrlOption set to swTwistControlConstantTwistAlongPath, then specify end twist angle

*BMergeSmoothFaces*

True to merge smooth faces, false to not

#### Return Value

Swept [body](#).

## Remarks

Use [IModelDocExtension::SelectByID2](#) to select the profile and sweep curves. The mark for:

- 1 = Profile selection
- 2 = Guide curve selection, if provided
- 4 = Sweep path

The PathAlign argument is available when twistCtrlOption is set to 0 (follow path) and can take one of these values:

- 0 = None; no correction (default)
- 2 = Direction vector; a plane, planar face, or line defines the path
- 3 = All faces; includes neighboring faces

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateBodiesFromSheets2 Method](#)  
[IModeler::CreateBodyFromBox Method](#)  
[IModeler::CreateBodyFromCone Method](#)  
[IModeler::CreateBodyFromCyl Method](#)  
[IModeler::CreateBodyFromFaces2 Method](#)  
[IModeler::CreateBrepBody3 Method](#)  
[IModeler::CreateExtrudedBody Method](#)  
[IModeler::CreateWireBody Method](#)  
[IModeler::ICreateBodiesFromSheets2 Method](#)  
[IModeler::ICreateBodyFromBox2 Method](#)  
[IModeler::ICreateBodyFromCone2 Method](#)  
[IModeler::ICreateBodyFromCyl2 Method](#)  
[IModeler::ICreateBodyFromFaces3 Method](#)  
[IModeler::ICreateBrepBody3 Method](#)  
[IModeler::ICreateWireBody Method](#)

## Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

## CreateSweptSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateSweptSurface Method (IModeler)

Creates a swept surface from a curve.

## .NET Syntax

Visual Basic (Declaration)

```
Function CreateSweptSurface( _  
    ByVal CurveIn As System.Object, _  
    ByVal Dir As System.Object _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IModeler  
Dim CurveIn As System.Object  
Dim Dir As System.Object  
Dim value As System.Object
```

```
value = instance.CreateSweptSurface(CurveIn, Dir)
```

C#

```
System.object CreateSweptSurface(  
    System.object CurveIn,  
    System.object Dir  
)
```

C++/CLI

```
System.Object^ CreateSweptSurface(  
& System.Object^ CurveIn,  
& System.Object^ Dir  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*CurveIn*

[Curve](#) to use to create the swept surface

*Dir*

3 doubles describing the direction vector to sweep the curve along (see **Remarks**)

#### Return Value

Newly created [surface](#)

### Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateSweptSurface](#).

### Remarks

This method creates an infinite surface. To create a trimmed surface, add a trimming loop with [ISurface::AddTrimmingLoop2](#) or [ISurface::!AddTrimmingLoop2](#).

The Dir array is:

- Dir[0] = X component of the direction vector
- Dir[1] = Y component of the direction vector
- Dir[2] = Z component of the direction vector

### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateBsplinesSurface Method](#)  
[IModeler::CreateConicalSurface2 Method](#)  
[IModeler::CreateCoonsBSurface Method](#)  
[IModeler::CreateCylindricalSurface2 Method](#)  
[IModeler::CreateExtrusionSurface Method](#)  
[IModeler::CreateLoftSurface Method](#)  
[IModeler::CreateOffsetSurface Method](#)  
[IModeler::CreatePlanarSurface2 Method](#)  
[IModeler::CreateRuledSurface Method](#)  
[IModeler::CreateToroidalSurface Method](#)  
[IModeler::!CreateBsplinesSurface Method](#)  
[IModeler::!CreateConicalSurface2 Method](#)  
[IModeler::!CreateCylindricalSurface2 Method](#)  
[IModeler::!CreateExtrusionSurface Method](#)  
[IModeler::!CreateLoftSurface Method](#)  
[IModeler::!CreateOffsetSurface Method](#)  
[IModeler::!CreatePlanarSurface2 Method](#)  
[IModeler::!CreateRuledSurface Method](#)  
[IModeler::!CreateSweptSurface Method](#)  
[IModeler::!CreateToroidalSurface Method](#)

### Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## CreateToroidalSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateToroidalSurface Method (IModeler)

Creates an untrimmed toroidal surface from the specified arguments.

### .NET Syntax

Visual Basic (Declaration)

```
Function CreateToroidalSurface( _  
    ByVal Center As System.Object, _  
    ByVal Axis As System.Object, _
```

```

    ByVal RefDirection As System.Object, _
    ByVal MajorRadius As System.Double, _
    ByVal MinorRadius As System.Double _
) As System.Object

```

Visual Basic (Usage)



```

Dim instance As IModeler
Dim Center As System.Object
Dim Axis As System.Object
Dim RefDirection As System.Object
Dim MajorRadius As System.Double
Dim MinorRadius As System.Double
Dim value As System.Object

value = instance.CreateToroidalSurface(Center, Axis, RefDirection, MajorRadius, MinorRadius)

```

C#

```

System.object CreateToroidalSurface(
    System.object Center,
    System.object Axis,
    System.object RefDirection,
    System.double MajorRadius,
    System.double MinorRadius
)

```

C++/CLI

```

System.Object^ CreateToroidalSurface(
    & System.Object^ Center,
    & System.Object^ Axis,
    & System.Object^ RefDirection,
    & System.double MajorRadius,
    & System.double MinorRadius
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Center*

Array containing 3 doubles (see **Remarks**)

*Axis*

Array containing 3 doubles (see **Remarks**)

*RefDirection*

Array containing 3 doubles (see **Remarks**)

*MajorRadius*

See **Remarks**

*MinorRadius*

See **Remarks**

#### Return Value

Untrimmed toroidal [surface](#)

### Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateToroidalSurface](#).

### Remarks

Input arguments:

- Center XYZ location that represents the center of the spherical surface.
- Axis XYZ direction of the axis of the spherical surface.
- RefDirection XYZ direction of the reference axis of the spherical surface; it must be perpendicular to the axis.
- MajorRadius major radius of the toroidal surface.
- MinorRadius minor radius of the toroidal surface.

The resulting surface can be trimmed using a method like [ISurface::CreateTrimmedSheet](#) to generate a sheet body.

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateConicalSurface2 Method](#)  
[IModeler::CreateCoonsBSurface Method](#)  
[IModeler::CreateCylindricalSurface2 Method](#)  
[IModeler::CreateExtrusionSurface Method](#)  
[IModeler::CreateLoftSurface Method](#)  
[IModeler::CreateOffsetSurface Method](#)  
[IModeler::CreatePlanarSurface2 Method](#)  
[IModeler::CreateRuledSurface Method](#)  
[IModeler::CreateSphericalSurface2 Method](#)  
[IModeler::CreateSweptSurface Method](#)  
[IModeler::ICreateB splineSurface Method](#)  
[IModeler::ICreateConicalSurface2 Method](#)  
[IModeler::ICreateCylindricalSurface2 Method](#)  
[IModeler::ICreateExtrusionSurface Method](#)  
[IModeler::ICreateLoftSurface Method](#)  
[IModeler::ICreateOffsetSurface Method](#)  
[IModeler::ICreatePlanarSurface2 Method](#)  
[IModeler::ICreateRuledSurface Method](#)  
[IModeler::ICreateSweptSurface Method](#)  
[IModeler::ICreateToroidalSurface Method](#)  
[ISurface::ISurface Method](#)  
[ISurface::TorusParams Property](#)

## Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments](#) on this topic.

## CreateWireBody Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateWireBody Method (IModeler)

Creates a wire body from the input entities, which can be edges or curves.

## .NET Syntax

### Visual Basic (Declaration)

```
Function CreateWireBody( _  
    ByVal EntVar As System.Object, _  
    ByVal Option As System.Integer _  
) As Body2
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim EntVar As System.Object  
Dim Option As System.Integer  
Dim value As Body2  
  
value = instance.CreateWireBody(EntVar, Option)
```

### C#

```
Body2 CreateWireBody(  
    System.object EntVar,  
    System.int Option  
)
```

### C++/CLI

```
Body2^ CreateWireBody(  
& System.Object^ EntVar,  
& System.int Option  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*EntVar*

Array of input entities ([curves](#) or [edges](#))

*Option*

Option as defined in [swCreateWireBodyOptions\\_e](#) (see **Remarks**)

#### Return Value

Newly created wire [body](#)

### Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateWireBody](#).

### Remarks

Wire bodies contain wires, loops, coedges, edges, and vertices. Wires typically represent profiles, construction lines, and center lines of swept shapes. Wires can also represent wire frames that, when surfaced, form shells.

The swCreateWireBodyMergeCurves option only works for curves with position and tangent continuity.

### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[ICurve::CreateWireBody Method](#)  
[IEdge::CreateWireBody Method](#)  
[IImportedCurveFeatureData::SetBody Method](#)  
[IModeler::CreateBodiesFromSheets2 Method](#)  
[IModeler::CreateBodyFromBox Method](#)  
[IModeler::CreateBodyFromCone Method](#)  
[IModeler::CreateBodyFromCyl Method](#)  
[IModeler::CreateBodyFromFaces2 Method](#)  
[IModeler::CreateBrepBody3 Method](#)  
[IModeler::CreateSweptBody Method](#)  
[IModeler::ICreateBodiesFromSheets2 Method](#)  
[IModeler::ICreateBodyFromBox2 Method](#)  
[IModeler::ICreateBodyFromCone2 Method](#)  
[IModeler::ICreateBodyFromCyl2 Method](#)  
[IModeler::ICreateBodyFromFaces3 Method](#)  
[IModeler::ICreateBrepBody3 Method](#)  
[IModeler::ICreateToroidalSurface Method](#)  
[IModeler::ICreateWireBody Method](#)  
[ISketchSegment::CreateWireBody Method](#)

### Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## DeleteFacesFromSheetBody Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : DeleteFacesFromSheetBody Method (IModeler)

Deletes the unused faces of the sheet body.

### .NET Syntax

Visual Basic (Declaration)

```
Function DeleteFacesFromSheetBody( _  
    ByVal FaceVar As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim FaceVar As System.Object  
Dim value As System.Boolean
```

```
value = instance.DeleteFacesFromSheetBody(FaceVar)
```

C#

```
System.bool DeleteFacesFromSheetBody(  
    System.object FaceVar  
)
```

```
System.bool DeleteFacesFromSheetBody(
& System.Object^ FaceVar
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*FaceVar*

Array of [faces](#) to delete from the sheet (surface) body

#### Visual Basic for Applications (VBA) Syntax

See [Modeler::DeleteFacesFromSheetBody](#).

#### Remarks

Use this method to remove the unused faces from the sheet body created by [IModeler::CreateBrepBody3](#) and [IModeler::!CreateBrepBody3](#).

#### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::!DeleteFacesFromSheetBody Method](#)

#### Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

## FindTwoEdgeMaxDeviation Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : FindTwoEdgeMaxDeviation Method (IModeler)

Finds the maximum distance between two edges.

#### .NET Syntax

##### Visual Basic (Declaration)

```
Function FindTwoEdgeMaxDeviation( _
    ByVal LpEdge1 As System.Object, _
    ByVal LpEdge2 As System.Object _
) As System.Double
```

##### Visual Basic (Usage)



```
Dim instance As IModeler
Dim LpEdge1 As System.Object
Dim LpEdge2 As System.Object
Dim value As System.Double
```

```
value = instance.FindTwoEdgeMaxDeviation(LpEdge1, LpEdge2)
```

##### C#

```
System.double FindTwoEdgeMaxDeviation(
    System.object LpEdge1,
    System.object LpEdge2
)
```

##### C++/CLI

```
System.double FindTwoEdgeMaxDeviation(
& System.Object^ LpEdge1,
```

```
& System.Object^ LpEdge2  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*LpEdge1*  
First [edge](#)

*LpEdge2*  
Second [edge](#)

#### Return Value

Maximum distance between the two edges

### Visual Basic for Applications (VBA) Syntax

See [Modeler::FindTwoEdgeMaxDeviation](#).

### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::IFindTwoEdgeMaxDeviation Method](#)

### Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments](#) on this topic.

## GeneralTopology Property (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : GeneralTopology Property (IModeler)

Gets or sets the Parasolid option to create general (non-manifold) bodies.

### .NET Syntax

#### Visual Basic (Declaration)

```
Property GeneralTopology As System.Boolean
```

#### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim value As System.Boolean  
  
instance.GeneralTopology = value  
  
value = instance.GeneralTopology
```

#### C#

```
System.bool GeneralTopology {get; set;}
```

#### C++/CLI

```
property System.bool GeneralTopology {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True to create general bodies, false to not

### Visual Basic for Applications (VBA) Syntax

See [Modeler::GeneralTopology](#).

## Example

[Create and Convert Non-manifold Bodies \(C#\)](#)  
[Create and Convert Non-manifold Bodies \(VB.NET\)](#)  
[Create and Convert Non-manifold Bodies \(VBA\)](#)

## Remarks

SOLIDWORKS expects this property be false because SOLIDWORKS does not support the creation of general bodies. If you set this property to True, then set it back to false as soon as your intended operations complete.

To tessellate a general body, set [ITessellation::MatchType](#) to swTessellationMatchFacetGeometry, which sets the Parasolid facet match to clipping facet boundaries to a common edge

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::GetManifoldBodiesCount Method](#)  
[IModeler::MakeManifoldBodies Method](#)  
[IModeler::MakeManifoldBodies Method](#)

## Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetBodyOutline Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : GetBodyOutline Method (IModeler)

Obsolete. Superseded by [IModeler::GetBodyOutline2](#).

## .NET Syntax

Visual Basic (Declaration)

```
Function GetBodyOutline( _  
    ByVal BodyVar As System.Object, _  
    ByVal Direction As MathVector, _  
    ByVal Tolerance As System.Double, _  
    ByRef CurvesOut As System.Object, _  
    ByRef TopolEntities As System.Object, _  
    ByRef Outline As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim BodyVar As System.Object  
Dim Direction As MathVector  
Dim Tolerance As System.Double  
Dim CurvesOut As System.Object  
Dim TopolEntities As System.Object  
Dim Outline As System.Object  
Dim value As System.Integer
```

```
value = instance.GetBodyOutline(BodyVar, Direction, Tolerance, CurvesOut, TopolEntities, Outline)
```

C#

```
System.int GetBodyOutline(  
    System.object BodyVar,  
    MathVector Direction,  
    System.double Tolerance,  
    out System.object CurvesOut,  
    out System.object TopolEntities,
```

```
    out System.Object Outline
)
```

C++/CLI

```
System.int GetBodyOutline(
& System.Object^ BodyVar,
& MathVector^ Direction,
& System.double Tolerance,
& [Out] System.Object^ CurvesOut,
& [Out] System.Object^ TopolEntities,
& [Out] System.Object^ Outline
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*BodyVar*

Array of bodies

*Direction*

Direction of view

*Tolerance*

Tolerance (Parasolid default 0.00001)

*CurvesOut*

Array of 3D trimmed curves that form the outline

*TopolEntities*

Array of topological entities associated with the outline

*Outline*

Array of integers indicating which curves belong to which outline

#### Return Value

Number of curves that form the outline of a body

### Visual Basic for Applications (VBA) Syntax

See [Modeler::GetBodyOutline](#).

### Remarks

See Parasolid's documentation of PK\_BODY\_make\_curves\_outline for more information about the output.

### See Also

[IModeler Interface](#)

[IModeler Members](#)

### Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetBodyOutline2 Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : GetBodyOutline2 Method (IModeler)

Gets the curves that form the outline of a body.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetBodyOutline2( _
    ByVal BodyList As System.Object, _
    ByVal Direction As MathVector, _
    ByVal Tolerance As System.Double, _
    ByVal ProjectOnPlane As System.Boolean, _
    ByRef Curves As System.Object, _
    ByRef TopolEntities As System.Object, _
```

```
ByRef Outline As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim BodyList As System.Object  
Dim Direction As MathVector  
Dim Tolerance As System.Double  
Dim ProjectOnPlane As System.Boolean  
Dim Curves As System.Object  
Dim TopolEntities As System.Object  
Dim Outline As System.Object  
Dim value As System.Integer  
  
value = instance.GetBodyOutline2(BodyList, Direction, Tolerance, ProjectOnPlane, Curves,  
TopolEntities, Outline)
```

C#

```
System.int GetBodyOutline2(  
    System.object BodyList,  
    MathVector Direction,  
    System.double Tolerance,  
    System.bool ProjectOnPlane,  
    out System.object Curves,  
    out System.object TopolEntities,  
    out System.object Outline  
)
```

C++/CLI

```
System.int GetBodyOutline2(  
& System.Object^ BodyList,  
& MathVector^ Direction,  
& System.double Tolerance,  
& System.bool ProjectOnPlane,  
& [Out] System.Object^ Curves,  
& [Out] System.Object^ TopolEntities,  
& [Out] System.Object^ Outline  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*BodyList*

Array of bodies

*Direction*

[IMathVector](#) direction of view

*Tolerance*

Tolerance (Parasolid default 0.00001)

*ProjectOnPlane*

True to project the curves onto a plane, false to not (see **Remarks**)

*Curves*

Array of 3D trimmed curves that form the outline

*TopolEntities*

Array of topological entities associated with the outline

*Outline*

Array of integers indicating which curves belong to which outline

#### Return Value

Number of curves that form the outline

## Visual Basic for Applications (VBA) Syntax

See [Modeler::GetBodyOutline2](#).

## Example

[Get Curves that Form Outline of Bodies \(VBA\)](#)

[Get Body Outline \(VBA\)](#)

[Get Body Outline \(VB.NET\)](#)  
[Get Body Outline \(C#\)](#)

## Remarks

When rendered, the array of curves returned in Curves might not display as a continuous outline of the body.

To close gaps:

1. Call this method with ProjectOnPlane = true.
2. Post-process the array of curves returned in Curves to interpolate the end points of each curve to generate a continuous outline of the body. See the **Get Body Outline** examples.

## See Also

[IModeler Interface](#)  
[IModeler Members](#)

## Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetInitKnitGapWidth Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : GetInitKnitGapWidth Method (IModeler)

Gets the initial gap bound width for knitting a body.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetInitKnitGapWidth() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim value As System.Double
```

```
value = instance.GetInitKnitGapWidth()
```

C#

```
System.double GetInitKnitGapWidth()
```

C++/CLI

```
System.double GetInitKnitGapWidth();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Initial knitting gap width used for body sewing

## Visual Basic for Applications (VBA) Syntax

See [Modeler::GetInitKnitGapWidth](#).

## Remarks

This value is 0 until a call to knit a body is made.

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::SetInitKnitGapWidth Method](#)

## GetManifoldBodiesCount Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : GetManifoldBodiesCount Method (IModeler)

Gets the number of manifold bodies created from the specified non-manifold body.

### .NET Syntax

#### Visual Basic (Declaration)

```
Function GetManifoldBodiesCount( _  
    ByVal NonManifoldIn As Body2 _  
) As System.Integer
```

#### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim NonManifoldIn As Body2  
Dim value As System.Integer
```

```
value = instance.GetManifoldBodiesCount(NonManifoldIn)
```

#### C#

```
System.int GetManifoldBodiesCount(  
    Body2 NonManifoldIn  
)
```

#### C++/CLI

```
System.int GetManifoldBodiesCount(  
& Body2^ NonManifoldIn  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*NonManifoldIn*  
Non-manifold [body](#)

#### Return Value

Number of manifold bodies

### Visual Basic for Applications (VBA) Syntax

See [Modeler::GetManifoldBodiesCount](#).

### Remarks

Call this method before calling [IModeler::IMakeManifoldBodies](#).

### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::GeneralTopology Property](#)  
[IModeler::MakeManifoldBodies Method](#)

### Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

## GetToleranceValue Method (IModeler)

[See Also](#) [Example](#)



Gets the current tolerance value of the given tolerance ID.

## Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetToleranceValue( _
    ByVal ToleranceID As System.Integer _
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim ToleranceID As System.Integer
Dim value As System.Double

value = instance.GetToleranceValue(ToleranceID)
```

C#

```
System.double GetToleranceValue(
    System.int ToleranceID
)
```

C++/CLI

```
System.double GetToleranceValue(
    & System.int ToleranceID
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*ToleranceID*

Type of tolerance that you want to obtain as defined by [swTolerances\\_e](#)

### Return Value

Tolerance value for the specified ToleranceID

## Visual Basic for Applications (VBA) Syntax

See [Modeler::GetToleranceValue](#).

## Example

[Get Length of Spline or Elliptical Edge \(VBA\)](#)  
[Get Parameters and Spline Points for Curve \(VBA\)](#)  
[Get Tolerance \(VBA\)](#)

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::UnsetTolerances Method](#)  
[IModeler::SetToleranceValue Method](#)

## Availability

SOLIDWORKS 2000 SP03, Revision Number 8.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

## ICheckInterference Method (IModeler)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > IModeler Interface : ICheckInterference Method (IModeler)

Obsolete. Superseded by [IModeler::ICheckInterference2](#).

## .NET Syntax

### Visual Basic (Declaration)

```
Sub ICheckInterference( _  
    ByRef Body1InterferedFaceArray As Face, _  
    ByRef Body2InterferedFaceArray As Face, _  
    ByRef IntersectedBodyArray As Body _  
)
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Body1InterferedFaceArray As Face  
Dim Body2InterferedFaceArray As Face  
Dim IntersectedBodyArray As Body  
  
instance.ICheckInterference(Body1InterferedFaceArray, Body2InterferedFaceArray,  
IntersectedBodyArray)
```

### C#

```
void ICheckInterference(  
    out Face Body1InterferedFaceArray,  
    out Face Body2InterferedFaceArray,  
    out Body IntersectedBodyArray  
)
```

### C++/CLI

```
void ICheckInterference(  
& [Out] Face^ Body1InterferedFaceArray,  
& [Out] Face^ Body2InterferedFaceArray,  
& [Out] Body^ IntersectedBodyArray  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Body1InterferedFaceArray*

*Body2InterferedFaceArray*

*IntersectedBodyArray*

## Visual Basic for Applications (VBA) Syntax

See [Modeler::ICheckInterference](#).

## See Also

[IModeler Interface](#)  
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

### ICheckInterference2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICheckInterference2 Method (IModeler)

Obsolete. Superseded by [IModeler::ICheckInterference3](#).

## .NET Syntax

### Visual Basic (Declaration)

```
Sub ICheckInterference2( _  
    ByRef Body1InterferedFaceArray As Face2, _  
    ByRef Body2InterferedFaceArray As Face2, _
```

```
ByRef IntersectedBodyArray As Body2 _  
)
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Body1InterferedFaceArray As Face2  
Dim Body2InterferedFaceArray As Face2  
Dim IntersectedBodyArray As Body2  
  
instance.ICheckInterference2(Body1InterferedFaceArray, Body2InterferedFaceArray,  
IntersectedBodyArray)
```

C#

```
void ICheckInterference2(  
    out Face2 Body1InterferedFaceArray,  
    out Face2 Body2InterferedFaceArray,  
    out Body2 IntersectedBodyArray  
)
```

C++/CLI

```
void ICheckInterference2(  
& [Out] Face2^ Body1InterferedFaceArray,  
& [Out] Face2^ Body2InterferedFaceArray,  
& [Out] Body2^ IntersectedBodyArray  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Body1InterferedFaceArray*  
Array of [faces](#) that interfered from the first body with the second body  
*Body2InterferedFaceArray*  
Array of [faces](#) that interfered from the second body with the first body  
*IntersectedBodyArray*  
[Bodies](#) generated from the intersection of the input bodies

### Visual Basic for Applications (VBA) Syntax

See [Modeler::ICheckInterference2](#).

### Remarks

Before calling this method, call [IModeler::ICheckInterferenceCount2](#) to allocate memory for the arrays returned from this method.

### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CheckInterference Method](#)  
[IAssemblyDoc::IToolsCheckInterference3 Method](#)  
[IAssemblyDoc::ToolsCheckInterference2 Method](#)

### Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## ICheckInterference3 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICheckInterference3 Method (IModeler)

Checks the interference among the specified bodies.

### .NET Syntax

Visual Basic (Declaration)

```
Function ICheckInterference3( _
    ByRef Body1InterferedFaceArray As Face2, _
    ByRef Body2InterferedFaceArray As Face2, _
    ByRef IntersectedBodyArray As Body2 _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim Body1InterferedFaceArray As Face2
Dim Body2InterferedFaceArray As Face2
Dim IntersectedBodyArray As Body2
Dim value As System.Boolean

value = instance.ICheckInterference3(Body1InterferedFaceArray, Body2InterferedFaceArray,
IntersectedBodyArray)
```

C#

```
System.bool ICheckInterference3(
    out Face2 Body1InterferedFaceArray,
    out Face2 Body2InterferedFaceArray,
    out Body2 IntersectedBodyArray
)
```

C++/CLI

```
System.bool ICheckInterference3(
    & [Out] Face2^ Body1InterferedFaceArray,
    & [Out] Face2^ Body2InterferedFaceArray,
    & [Out] Body2^ IntersectedBodyArray
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Body1InterferedFaceArray*  
Array of [faces](#) that have interfered from the first body with the second body  
*Body2InterferedFaceArray*  
Array of [faces](#) that interfered from the second body with the first body  
*IntersectedBodyArray*  
Array of interfering [bodies](#)

#### Return Value

True if a clash occurred, false if not

### Visual Basic for Applications (VBA) Syntax

See [Modeler::ICheckInterference3](#).

### Remarks

Before calling this method, call [IModeler::ICheckInterferenceCount3](#) to allocate memory for the arrays returned from this method.

### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IAssemblyDoc::IToolsCheckInterference3 Method](#)  
[IAssemblyDoc::ToolsCheckInterference2 Method](#)  
[IModeler::CheckInterference3 Method](#)  
[IModeler::ICheckInterferenceCount3 Method](#)

### Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0



Obsolete. Superseded by [IModeler::ICheckInterferenceCount2](#).

## Visual Basic Syntax

### Visual Basic (Declaration)

```
Function ICheckInterferenceCount( _  
    ByVal Body1 As Body, _  
    ByVal Body2 As Body, _  
    ByVal CoincidentInterference As System.Boolean, _  
    ByRef Body1InterferedFaceCount As System.Integer, _  
    ByRef Body2InterferedFaceCount As System.Integer, _  
    ByRef IntersectedBodyCount As System.Integer _  
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Body1 As Body  
Dim Body2 As Body  
Dim CoincidentInterference As System.Boolean  
Dim Body1InterferedFaceCount As System.Integer  
Dim Body2InterferedFaceCount As System.Integer  
Dim IntersectedBodyCount As System.Integer  
Dim value As System.Boolean  
  
value = instance.ICheckInterferenceCount(Body1, Body2, CoincidentInterference,  
Body1InterferedFaceCount, Body2InterferedFaceCount, IntersectedBodyCount)
```

### C#

```
System.bool ICheckInterferenceCount(  
    Body Body1,  
    Body Body2,  
    System.bool CoincidentInterference,  
    out System.int Body1InterferedFaceCount,  
    out System.int Body2InterferedFaceCount,  
    out System.int IntersectedBodyCount  
)
```

### C++/CLI

```
System.bool ICheckInterferenceCount(  
& Body^ Body1,  
& Body^ Body2,  
& System.bool CoincidentInterference,  
& [Out] System.int Body1InterferedFaceCount,  
& [Out] System.int Body2InterferedFaceCount,  
& [Out] System.int IntersectedBodyCount  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*Body1*

*Body2*

*CoincidentInterference*

*Body1InterferedFaceCount*

*Body2InterferedFaceCount*

*IntersectedBodyCount*

## Visual Basic for Applications (VBA) Syntax

See [Modeler::ICheckInterferenceCount](#).

## See Also

[IModeler Interface](#)  
[IModeler Members](#)

SOLIDWORKS API Help

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## ICheckInterferenceCount2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICheckInterferenceCount2 Method (IModeler)

Obsolete. Superseded by [IModeler::ICheckInterferenceCount3](#).

## .NET Syntax

### Visual Basic (Declaration)

```
Function ICheckInterferenceCount2( _  
    ByVal Body1 As Body2, _  
    ByVal Body2 As Body2, _  
    ByVal CoincidentInterference As System.Boolean, _  
    ByRef Body1InterferedFaceCount As System.Integer, _  
    ByRef Body2InterferedFaceCount As System.Integer, _  
    ByRef IntersectedBodyCount As System.Integer _  
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Body1 As Body2  
Dim Body2 As Body2  
Dim CoincidentInterference As System.Boolean  
Dim Body1InterferedFaceCount As System.Integer  
Dim Body2InterferedFaceCount As System.Integer  
Dim IntersectedBodyCount As System.Integer  
Dim value As System.Boolean  
  
value = instance.ICheckInterferenceCount2(Body1, Body2, CoincidentInterference,  
Body1InterferedFaceCount, Body2InterferedFaceCount, IntersectedBodyCount)
```

### C#

```
System.bool ICheckInterferenceCount2(  
    Body2 Body1,  
    Body2 Body2,  
    System.bool CoincidentInterference,  
    out System.int Body1InterferedFaceCount,  
    out System.int Body2InterferedFaceCount,  
    out System.int IntersectedBodyCount  
)
```

### C++/CLI

```
System.bool ICheckInterferenceCount2(  
& Body2^ Body1,  
& Body2^ Body2,  
& System.bool CoincidentInterference,  
& [Out] System.int Body1InterferedFaceCount,  
& [Out] System.int Body2InterferedFaceCount,  
& [Out] System.int IntersectedBodyCount  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*Body1*  
First [body](#) to check for interference  
*Body2*  
Second [body](#) to check for interference  
*CoincidentInterference*  
True to check for coincident interference, false to not  
*Body1InterferedFaceCount*

Number of faces that are interfering that belong to the body passed in the first parameter of this method  
*Body2InterferedFaceCount*  
Number of faces that are interfering that belong to the body passed in the second parameter of this method  
*IntersectedBodyCount*  
Number of intersection bodies produced from this intersection

#### Return Value

True if successful, false if not

### Visual Basic for Applications (VBA) Syntax

See [Modeler::ICheckInterferenceCount2](#).

### Remarks

Call [IModeler::ICheckInterference2](#) after calling this method.

### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CheckInterference Method](#)  
[IAssemblyDoc::ToolsCheckInterference2 Method](#)  
[IAssemblyDoc::ToolsCheckInterference3 Method](#)

### Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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## ICheckInterferenceCount3 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICheckInterferenceCount3 Method (IModeler)

Checks interference among the specified bodies and returns the number of interferences.

### .NET Syntax

Visual Basic (Declaration)

```
Function ICheckInterferenceCount3( _  
    ByVal NumOfTargetBodies As System.Integer, _  
    ByRef TargetBodies As Body2, _  
    ByVal NumOfToolBodies As System.Integer, _  
    ByRef ToolBodies As Body2, _  
    ByVal Options As System.Integer, _  
    ByRef NumBody1InterferedFaceArray As System.Integer, _  
    ByRef NumBody2InterferedFaceArray As System.Integer, _  
    ByRef NumIntersectedBodyArray As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim NumOfTargetBodies As System.Integer  
Dim TargetBodies As Body2  
Dim NumOfToolBodies As System.Integer  
Dim ToolBodies As Body2  
Dim Options As System.Integer  
Dim NumBody1InterferedFaceArray As System.Integer  
Dim NumBody2InterferedFaceArray As System.Integer  
Dim NumIntersectedBodyArray As System.Integer  
Dim value As System.Boolean
```

```
value = instance.ICheckInterferenceCount3(NumOfTargetBodies, TargetBodies, NumOfToolBodies,  
ToolBodies, Options, NumBody1InterferedFaceArray, NumBody2InterferedFaceArray,  
NumIntersectedBodyArray)
```

C#

```

System.bool ICheckInterferenceCount3(
    System.int NumOfTargetBodies,
    ref Body2 TargetBodies,
    System.int NumOfToolBodies,
    ref Body2 ToolBodies,
    System.int Options,
    out System.int NumBody1InterferedFaceArray,
    out System.int NumBody2InterferedFaceArray,
    out System.int NumIntersectedBodyArray
)

```

C++/CLI

```

System.bool ICheckInterferenceCount3(
    & System.int NumOfTargetBodies,
    & Body2^% TargetBodies,
    & System.int NumOfToolBodies,
    & Body2^% ToolBodies,
    & System.int Options,
    & [Out] System.int NumBody1InterferedFaceArray,
    & [Out] System.int NumBody2InterferedFaceArray,
    & [Out] System.int NumIntersectedBodyArray
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*NumOfTargetBodies*  
Number of target bodies  
*TargetBodies*

Array of target [bodies](#)

*NumOfToolBodies*  
Number of tool bodies  
*ToolBodies*

Array of tool [bodies](#)

*Options*  
Check interference options as defined by [swCheckInterferenceOption\\_e](#)  
*NumBody1InterferedFaceArray*  
Number of [faces](#) that interfered from the first body with the second body  
*NumBody2InterferedFaceArray*  
Number of [faces](#) that interfered from the second body with the first body  
*NumIntersectedBodyArray*  
Number of interfering [bodies](#)

#### Visual Basic for Applications (VBA) Syntax

See [Modeler::ICheckInterferenceCount3](#).

#### Remarks

Call this method before calling [IModeler::ICheckInterference3](#) to get the size of the arrays for that method.

#### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::ICheckInterference3 Method](#)  
[IModeler::CheckInterference3 Method](#)  
[IAssemblyDoc::IToolsCheckInterference3 Method](#)  
[IAssemblyDoc::ToolsCheckInterference2 Method](#)

#### Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## ICopyWizardHole Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICopyWizardHole Method (IModeler)

Copies hole data from the source hole to the destination hole.

## Visual Basic Syntax

### Visual Basic (Declaration)

```
Function ICopyWizardHole( _  
    ByVal SourceHole As Feature, _  
    ByVal DestinationHole As Feature, _  
    ByVal RebuildOwner As System.Boolean _  
) As System.Integer
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim SourceHole As Feature  
Dim DestinationHole As Feature  
Dim RebuildOwner As System.Boolean  
Dim value As System.Integer  
  
value = instance.ICopyWizardHole(SourceHole, DestinationHole, RebuildOwner)
```

### C#

```
System.int ICopyWizardHole(  
    Feature SourceHole,  
    Feature DestinationHole,  
    System.bool RebuildOwner  
)
```

### C++/CLI

```
System.int ICopyWizardHole(  
& Feature^ SourceHole,  
& Feature^ DestinationHole,  
& System.bool RebuildOwner  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*SourceHole*  
[Source hole](#)  
*DestinationHole*  
[Destination hole](#)  
*RebuildOwner*

True rebuilds the model, false does not

#### Return Value

0 if the call generated an error; 1 if the call did not generate an error

## Visual Basic for Applications (VBA) Syntax

See [Modeler::ICopyWizardHole](#).

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler.CopyWizardHole Method](#)  
[IWizardHoleFeatureData2 Interface](#)

## Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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## CreateArc Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateArc Method (IModeler)

Creates an arc.

## .NET Syntax

### Visual Basic (Declaration)

```
Function ICreateArc( _  
    ByRef Center As System.Double, _  
    ByRef Axis As System.Double, _  
    ByVal Radius As System.Double, _  
    ByRef StartPoint As System.Double, _  
    ByRef EndPoint As System.Double _  
) As Curve
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Double  
Dim Axis As System.Double  
Dim Radius As System.Double  
Dim StartPoint As System.Double  
Dim EndPoint As System.Double  
Dim value As Curve  
  
value = instance.ICreateArc(Center, Axis, Radius, StartPoint, EndPoint)
```

### C#

```
Curve ICreateArc(  
    ref System.double Center,  
    ref System.double Axis,  
    System.double Radius,  
    ref System.double StartPoint,  
    ref System.double EndPoint  
)
```

### C++/CLI

```
Curve^ ICreateArc(  
& System.double% Center,  
& System.double% Axis,  
& System.double% Radius,  
& System.double% StartPoint,  
& System.double% EndPoint  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Center*  
Array containing 3 doubles for location of the center of the arc (x,y,z)  
*Axis*  
Array containing 3 doubles (x,y,z)  
*Radius*  
Arc radius  
*StartPoint*  
Array of 3 doubles (x,y,z)  
*EndPoint*  
Array of 3 doubles (x,y,z)

#### Return Value

[Curve](#)

## Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateArc](#).

## Remarks

If your application uses [ISurface::CreateTrimmedSheet4](#) or [ISurface::ICreateTrimmedSheet4](#), then your application must also use [ICurve::CreateTrimmedCurve2](#) for the curves created by [IModeler::CreateArc](#) or [IModeler::ICreateArc](#) and [IModeler::CreateLine](#) or [IModeler::ICreateLine](#).

## See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IBody2::AddProfileArc Method](#)  
[IBody2::IAddProfileArc Method](#)  
[IModeler::CreateEllipse Method](#)  
[IModeler::ICreateEllipse Method](#)

SOLIDWORKS API Help

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## ICreateBodiesFromSheets Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBodiesFromSheets Method (IModeler)

Obsolete. Superseded by [IModeler::ICreateBodiesFromSheets2](#).

## .NET Syntax

### Visual Basic (Declaration)

```
Function ICreateBodiesFromSheets( _  
    ByVal NSheets As System.Integer, _  
    ByRef Sheets As System.Object, _  
    ByVal Options As System.Integer, _  
    ByRef NResults As System.Integer, _  
    ByRef Results As System.Object _  
    ) As System.Integer
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim NSheets As System.Integer  
Dim Sheets As System.Object  
Dim Options As System.Integer  
Dim NResults As System.Integer  
Dim Results As System.Object  
Dim value As System.Integer  
  
value = instance.ICreateBodiesFromSheets(NSheets, Sheets, Options, NResults, Results)
```

### C#

```
System.int ICreateBodiesFromSheets(  
    System.int NSheets,  
    ref System.object Sheets,  
    System.int Options,  
    out System.int NResults,  
    out System.object Results  
)
```

### C++/CLI

```
System.int ICreateBodiesFromSheets(  
& System.int NSheets,  
& System.Object^% Sheets,  
& System.int Options,  
& [Out] System.int NResults,  
& [Out] System.Object^ Results  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*NSheets*

*Sheets*

*Options*

*NResults*

*Results*

## Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBodiesFromSheets](#).

## See Also

[IModeler Interface](#)  
[IModeler Members](#)

SOLIDWORKS API Help

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## ICreateBodiesFromSheets2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBodiesFromSheets2 Method (IModeler)

Sews sheets to make a sheet (surface) or solid bodies.

## .NET Syntax

### Visual Basic (Declaration)

```
Function ICreateBodiesFromSheets2( _  
    ByVal NSheets As System.Integer, _  
    ByRef Sheets As Body2, _  
    ByVal Options As System.Integer, _  
    ByVal KnittingTolerance As System.Double, _  
    ByRef NResults As System.Integer, _  
    ByRef Results As Body2 _  
) As System.Integer
```

### Visual Basic (Usage)



```
Dim instance As IModeler  
Dim NSheets As System.Integer  
Dim Sheets As Body2  
Dim Options As System.Integer  
Dim KnittingTolerance As System.Double  
Dim NResults As System.Integer  
Dim Results As Body2  
Dim value As System.Integer
```

```
value = instance.ICreateBodiesFromSheets2(NSheets, Sheets, Options, KnittingTolerance, NResults, Results)
```

### C#

```
System.int ICreateBodiesFromSheets2(  
    System.int NSheets,  
    ref Body2 Sheets,  
    System.int Options,  
    System.double KnittingTolerance,  
    out System.int NResults,  
    out Body2 Results  
)
```

### C++/CLI

```
System.int ICreateBodiesFromSheets2(  
& System.int NSheets,  
& Body2^% Sheets,  
& System.int Options,  
& System.double KnittingTolerance,  
& [Out] System.int NResults,  
& [Out] Body2^ Results  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Parameters

*NSheets*  
Number of sheets

*Sheets*  
Array containing the [sheets](#)  
*Options*  
Type of body to create as defined by [swSheetSewingOption\\_e](#)  
*KnittingTolerance*  
Knitting tolerance  
*NResults*  
Number of sheet or solid bodies created  
*Results*  
Array of the sheet or solid [bodies](#)

**Return Value**

Error as defined by [swSheetSewingError\\_e](#)

 **Visual Basic for Applications (VBA) Syntax**

See [Modeler::ICreateBodiesFromSheets2](#).

 **Remarks**

It is safest to assume that the maximum number of bodies that can be returned by this method is the number of sheet input bodies. Your code should look like this:

```
aBodies = new IBody2*[lNumSheetBodies];  
  
hres = swModeler->ICreateBodiesFromSheets2(lNumSheetBodies, aSheetBodies, swSewToSheets, 1e-6,  
&lNumBodies, aBodies, &lErrors);
```

 **See Also**

- [IModeler Interface](#)
- [IModeler Members](#)
- [IModeler::CreateBodiesFromSheets2 Method](#)
- [IModeler::CreateBodyFromBox Method](#)
- [IModeler::CreateBodyFromCone Method](#)
- [IModeler::CreateBodyFromCyl Method](#)
- [IModeler::CreateBodyFromFaces2 Method](#)
- [IModeler::CreateBrepBody3 Method](#)
- [IModeler::CreateExtrudedBody Method](#)
- [IModeler::CreateSweptBody Method](#)
- [IModeler::CreateWireBody Method](#)
- [IModeler::ICreateBodyFromBox2 Method](#)
- [IModeler::ICreateBodyFromCone2 Method](#)
- [IModeler::ICreateBodyFromCyl2 Method](#)
- [IModeler::ICreateBodyFromFaces3 Method](#)
- [IModeler::ICreateBrepBody3 Method](#)
- [IModeler::ICreateWireBody Method](#)

 **Availability**

SOLIDWORKS 2007 FCS, Revision Number 15.0

**ICreateBodyFromBox Method (IModeler)**

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBodyFromBox Method (IModeler)

Obsolete. Superseded by [IModeler::ICreateBodyFromBox2](#)

 **.NET Syntax**

Visual Basic (Declaration)

```
Function ICreateBodyFromBox( _  
    ByRef BoxDimArray As System.Double _  
) As Body
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim BoxDimArray As System.Double
```

```
Dim value As Body

value = instance.ICreateBodyFromBox(BoxDimArray)
```

C#

```
Body ICreateBodyFromBox(
    ref System.double BoxDimArray
)
```

C++/CLI

```
Body^ ICreateBodyFromBox(
    & System.double% BoxDimArray
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*BoxDimArray*

### Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBodyFromBox](#).

### See Also

[IModeler Interface](#)  
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## ICreateBodyFromBox2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBodyFromBox2 Method (IModeler)

Obsolete. Superseded by [IModeler::CreateBodyFromBox3](#).

### .NET Syntax

Visual Basic (Declaration)

```
Function ICreateBodyFromBox2( _
    ByRef BoxDimArray As System.Double _
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim BoxDimArray As System.Double
Dim value As Body2
```

```
value = instance.ICreateBodyFromBox2(BoxDimArray)
```

C#

```
Body2 ICreateBodyFromBox2(
    ref System.double BoxDimArray
)
```

C++/CLI

```
Body2^ ICreateBodyFromBox2(
    & System.double% BoxDimArray
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

BoxDimArray  
Array of 9 doubles (see **Remarks**)

Return Value

[Body](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBodyFromBox2](#).

Remarks

The input parameter is the following array of doubles:

[ *boxFaceCenter*[3], *boxAxis*[3], *boxWidth*, *boxLength*, *boxHeight* ]

where:

|                          |                                                                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>boxFaceCenter</i> [3] | XYZ location that represents the center of one of the box faces.                                                                                                                                             |
| <i>boxAxis</i> [3]       | XYZ direction. The box will be extruded along this vector from the <i>boxFaceCenter</i> location, a distance of <i>boxHeight</i> .                                                                           |
| <i>boxWidth</i>          | Box width. If <i>boxAxis</i> is parallel to the Z axis (0,0,1), then this value represents the dimension that is parallel to the X-axis; otherwise, the orientation is not defined.                          |
| <i>boxLength</i>         | Box length. If <i>boxAxis</i> is parallel to the Z axis (0,0,1), then this value represents the dimension that is parallel to the Y axis; otherwise, the orientation is not defined.                         |
| <i>boxHeight</i>         | Height to extrude along the <i>boxAxis</i> direction. If <i>boxHeight</i> is 0, a sheet body will be created of dimension <i>boxWidth</i> x <i>boxLength</i> and whose normal is defined by <i>boxAxis</i> . |

See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateBodiesFromSheets2 Method](#)  
[IModeler::CreateBodyFromBox Method](#)  
[IModeler::CreateBodyFromCone Method](#)  
[IModeler::CreateBodyFromCyl Method](#)  
[IModeler::CreateBodyFromFaces2 Method](#)  
[IModeler::CreateBrepBody3 Method](#)  
[IModeler::CreateExtrudedBody Method](#)  
[IModeler::CreateSweptBody Method](#)  
[IModeler::CreateToroidalSurface Method](#)  
[IModeler::CreateWireBody Method](#)  
[IModeler::ICreateBodiesFromSheets2 Method](#)  
[IModeler::ICreateBodyFromCone2 Method](#)  
[IModeler::ICreateBodyFromCyl2 Method](#)  
[IModeler::ICreateBodyFromFaces3 Method](#)  
[IModeler::ICreateBrepBody3 Method](#)  
[IModeler::ICreateToroidalSurface Method](#)  
[IModeler::ICreateWireBody Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateBodyFromCone Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBodyFromCone Method (IModeler)

Obsolete. Superseded by [IModeler::ICreateBodyFromCone2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateBodyFromCone( _  
    ByRef ConeDimArray As System.Double _  
) As Body
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim ConeDimArray As System.Double
Dim value As Body

value = instance.ICreateBodyFromCone(ConeDimArray)
```

C#

```
Body ICreateBodyFromCone(
    ref System.double ConeDimArray
)
```

C++/CLI

```
Body^ ICreateBodyFromCone(
    & System.double% ConeDimArray
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*ConeDimArray*

### Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBodyFromCone](#).

### See Also

[IModeler Interface](#)  
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## ICreateBodyFromCone2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBodyFromCone2 Method (IModeler)

Creates a temporary body from cone dimensions.

### .NET Syntax

Visual Basic (Declaration)

```
Function ICreateBodyFromCone2( _
    ByRef ConeDimArray As System.Double _
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim ConeDimArray As System.Double
Dim value As Body2
```

```
value = instance.ICreateBodyFromCone2(ConeDimArray)
```

C#

```
Body2 ICreateBodyFromCone2(
    ref System.double ConeDimArray
)
```

C++/CLI

```
Body2^ ICreateBodyFromCone2(
    & System.double% ConeDimArray
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

**Parameters**

*ConeDimArray*  
Array of 9 doubles (see **Remarks**)

**Return Value**

[Body](#).

 **Visual Basic for Applications (VBA) Syntax**

See [Modeler::ICreateBodyFromCone2](#).

 **Remarks**

The ConeArray argument is the following array of doubles:

[ *coneFaceCenter*[3], *coneAxis*[3], *coneBaseRadius*, *coneTopRadius*, *coneHeight* ]

where:

|                           |                                                                                                                                                                                            |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>coneFaceCenter</i> [3] | XYZ location that represents the center of one of the cone faces                                                                                                                           |
| <i>coneAxis</i> [3]       | XYZ direction; the cone is extruded along this vector from the <i>coneFaceCenter</i> location, a distance of <i>coneHeight</i>                                                             |
| <i>coneBaseRadius</i>     | Cone radius at the <i>coneFaceCenter</i> plane                                                                                                                                             |
| <i>coneTopRadius</i>      | Cone radius at <i>coneHeight</i> from the <i>coneFaceCenter</i> along the axis                                                                                                             |
| <i>coneHeight</i>         | Length to extrude along the <i>coneAxis</i> direction; if <i>coneHeight</i> is 0, then a sheet body is created of dimension <i>coneBaseRadius</i> and normal is defined by <i>coneAxis</i> |

 **See Also**

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateBodiesFromSheets2 Method](#)  
[IModeler::CreateBodyFromBox Method](#)  
[IModeler::CreateBodyFromCone Method](#)  
[IModeler::CreateBodyFromCyl Method](#)  
[IModeler::CreateBodyFromFaces2 Method](#)  
[IModeler::CreateBrepBody3 Method](#)  
[IModeler::CreateExtrudedBody Method](#)  
[IModeler::CreateSweptBody Method](#)  
[IModeler::CreateWireBody Method](#)  
[IModeler::ICreateBodiesFromSheets2 Method](#)  
[IModeler::ICreateBodyFromBox2 Method](#)  
[IModeler::ICreateBodyFromCyl2 Method](#)  
[IModeler::ICreateBodyFromFaces3 Method](#)  
[IModeler::ICreateBrepBody3 Method](#)  
[IModeler::ICreateWireBody Method](#)

 **Availability**

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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**ICreateBodyFromCyl Method (IModeler)**

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : [ICreateBodyFromCyl Method \(IModeler\)](#)

**Obsolete.** Superseded by [IModeler::ICreateBodyFromCyl2](#).

 **.NET Syntax**

Visual Basic (Declaration)

```
Function ICreateBodyFromCyl( _  
    ByRef CylDimArray As System.Double _  
) As Body
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim CylDimArray As System.Double  
Dim value As Body  
  
value = instance.ICreateBodyFromCyl(CylDimArray)
```

C#

```
Body ICreateBodyFromCyl(  
    ref System.double CylDimArray  
)
```

C++/CLI

```
Body^ ICreateBodyFromCyl(  
& System.double% CylDimArray  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*CylDimArray*

### Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBodyFromCyl](#).

### See Also

[IModeler Interface](#)  
[IModeler Members](#)

SOLIDWORKS API Help

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## ICreateBodyFromCyl2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBodyFromCyl2 Method (IModeler)

Creates a temporary body from cylinder dimensions.

### .NET Syntax

Visual Basic (Declaration)

```
Function ICreateBodyFromCyl2( _  
    ByRef CylDimArray As System.Double _  
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim CylDimArray As System.Double  
Dim value As Body2
```

```
value = instance.ICreateBodyFromCyl2(CylDimArray)
```

C#

```
Body2 ICreateBodyFromCyl2(  
    ref System.double CylDimArray  
)
```

C++/CLI

```
Body2^ ICreateBodyFromCyl2(  
& System.double% CylDimArray  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

**Parameters**

*CylDimArray*  
Array containing 8 doubles (see **Remarks**)

 **Visual Basic for Applications (VBA) Syntax**

See [Modeler::ICreateBodyFromCyl2](#).

 **Remarks**

The *CylArray* argument is the following array of doubles:

[ *cylFaceCenter*[3], *cylAxis*[3], *cylRadius*, *cylHeight* ]

where:

|                          |                                                                                                                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>cylFaceCenter</i> [3] | XYZ location that represents the center of one of the cylinder faces                                                                                                                     |
| <i>cylAxis</i> [3]       | XYZ direction; the cylinder is extruded along this vector from the <i>cylFaceCenter</i> location for a distance of <i>cylHeight</i>                                                      |
| <i>cylRadius</i>         | Cylinder radius                                                                                                                                                                          |
| <i>cylHeight</i>         | Length to extrude along the <i>cylAxis</i> direction; if <i>cylHeight</i> is 0, then a sheet body is created of dimension <i>cylRadius</i> and whose normal is defined by <i>cylAxis</i> |

 **See Also**

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateBodiesFromSheets2 Method](#)  
[IModeler::CreateBodyFromBox Method](#)  
[IModeler::CreateBodyFromCone Method](#)  
[IModeler::CreateBodyFromFaces2 Method](#)  
[IModeler::CreateBrepBody3 Method](#)  
[IModeler::CreateExtrudedBody Method](#)  
[IModeler::CreateSweptBody Method](#)  
[IModeler::CreateWireBody Method](#)  
[IModeler::ICreateBodiesFromSheets2 Method](#)  
[IModeler::ICreateBodyFromBox2 Method](#)  
[IModeler::ICreateBodyFromCone2 Method](#)  
[IModeler::ICreateBodyFromFaces3 Method](#)  
[IModeler::ICreateBrepBody3 Method](#)  
[IModeler::ICreateWireBody Method](#)

 **Availability**

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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**ICreateBodyFromFaces Method (IModeler)**

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBodyFromFaces Method (IModeler)

Obsolete. Superseded by [IModeler::ICreateBodyFromFace3](#).

 **.NET Syntax**

Visual Basic (Declaration)

```
Function ICreateBodyFromFaces( _  
    ByVal NumOfFaces As System.Integer, _  
    ByRef Faces As Face, _  
    ByVal DoLocalCheck As System.Boolean, _  
    ByRef LocalCheckResult As System.Boolean _  
    ) As Body
```

Visual Basic (Usage)



```

Dim instance As IModeler
Dim NumOfFaces As System.Integer
Dim Faces As Face
Dim DoLocalCheck As System.Boolean
Dim LocalCheckResult As System.Boolean
Dim value As Body

value = instance.ICreateBodyFromFaces(NumOfFaces, Faces, DoLocalCheck, LocalCheckResult)

```

C#

```

Body ICreateBodyFromFaces(
    System.int NumOfFaces,
    ref Face Faces,
    System.bool DoLocalCheck,
    ref System.bool LocalCheckResult
)

```

C++/CLI

```

Body^ ICreateBodyFromFaces(
    & System.int NumOfFaces,
    & Face^% Faces,
    & System.bool DoLocalCheck,
    & System.bool% LocalCheckResult
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*NumOfFaces*

*Faces*

*DoLocalCheck*

*LocalCheckResult*

### Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBodyFromFaces](#).

### See Also

[IModeler Interface](#)  
[IModeler Members](#)

SOLIDWORKS API Help

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## ICreateBodyFromFaces2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBodyFromFaces2 Method (IModeler)

Obsolete. Superseded by [IModeler::ICreateBodyFromFace3](#).

### .NET Syntax

Visual Basic (Declaration)

```

Function ICreateBodyFromFaces2( _
    ByVal NumOfFaces As System.Integer, _
    ByRef Faces As Face, _
    ByVal ActionType As System.Integer, _
    ByVal DoLocalCheck As System.Boolean, _
    ByRef LocalCheckResult As System.Boolean _
) As Body

```

Visual Basic (Usage)

 Copy Code

```

Dim instance As IModeler
Dim NumOfFaces As System.Integer
Dim Faces As Face
Dim ActionType As System.Integer
Dim DoLocalCheck As System.Boolean
Dim LocalCheckResult As System.Boolean
Dim value As Body

```

```

value = instance.ICreateBodyFromFaces2(NumOfFaces, Faces, ActionType, DoLocalCheck,
LocalCheckResult)

```

C#

```

Body ICreateBodyFromFaces2(
    System.int NumOfFaces,
    ref Face Faces,
    System.int ActionType,
    System.bool DoLocalCheck,
    out System.bool LocalCheckResult
)

```

C++/CLI

```

Body^ ICreateBodyFromFaces2(
    & System.int NumOfFaces,
    & Face^% Faces,
    & System.int ActionType,
    & System.bool DoLocalCheck,
    [Out] System.bool LocalCheckResult
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*NumOfFaces*

*Faces*

*ActionType*

*DoLocalCheck*

*LocalCheckResult*

### Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBodyFromFaces2](#).

### See Also

[IModeler Interface](#)

[IModeler Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## ICreateBodyFromFaces3 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBodyFromFaces3 Method (IModeler)

Creates a temporary body from a list of faces.

### .NET Syntax

Visual Basic (Declaration)

```

Function ICreateBodyFromFaces3( _
    ByVal NumOfFaces As System.Integer, _
    ByRef Faces As Face2, _
    ByVal ActionType As System.Integer, _
    ByVal DoLocalCheck As System.Boolean, _

```

```
ByRef LocalCheckResult As System.Boolean _  
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim NumOfFaces As System.Integer  
Dim Faces As Face2  
Dim ActionType As System.Integer  
Dim DoLocalCheck As System.Boolean  
Dim LocalCheckResult As System.Boolean  
Dim value As Body2  
  
value = instance.ICreateBodyFromFaces3(NumOfFaces, Faces, ActionType, DoLocalCheck,  
LocalCheckResult)
```

C#

```
Body2 ICreateBodyFromFaces3(  
    System.int NumOfFaces,  
    ref Face2 Faces,  
    System.int ActionType,  
    System.bool DoLocalCheck,  
    out System.bool LocalCheckResult  
)
```

C++/CLI

```
Body2^ ICreateBodyFromFaces3(  
& System.int NumOfFaces,  
& Face2^% Faces,  
& System.int ActionType,  
& System.bool DoLocalCheck,  
& [Out] System.bool LocalCheckResult  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*NumOfFaces*

Number of faces to use to create the body

*Faces*

Array of [faces](#)

*ActionType*

Type of action as defined in [swCreateFacesBodyAction\\_e](#)

*DoLocalCheck*

True to perform local checking on the resulting body, false to not

*LocalCheckResult*

If DoLocalCheck is True, then True if body is okay, false if not

#### Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBodyFromFaces3](#).

#### Remarks

This method is useful when you attempt to copy a body with changes to that body. All of the input faces must be from the same body.

If you call this method to remove a hole, then the value that you specify for ActionType indicates to the modeler how to handle filling the hole. The result must be a solid.

This method is the opposite of [IBody2::DeleteFaces3](#) or [IBody2::DeleteFaces3](#), which allows you to specify faces to remove.

#### See Also

[IModeler Interface](#)  
[IModeler Members](#)  
[IModeler::CreateBodiesFromSheets2 Method](#)  
[IModeler::CreateBodyFromBox Method](#)  
[IModeler::CreateBodyFromCone Method](#)  
[IModeler::CreateBodyFromCyl Method](#)  
[IModeler::CreateBodyFromFaces2 Method](#)

[IModeler::CreateBrepBody3 Method](#)  
[IModeler::CreateExtrudedBody Method](#)  
[IModeler::CreateSweepBody Method](#)  
[IModeler::CreateWireBody Method](#)  
[IModeler::ICreateBodiesFromSheets2 Method](#)  
[IModeler::ICreateBodyFromBox2 Method](#)  
[IModeler::ICreateBodyFromCone2 Method](#)  
[IModeler::ICreateBodyFromCyl2 Method](#)  
[IModeler::ICreateBrepBody3 Method](#)  
[IModeler::ICreateWireBody Method](#)

## Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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## ICreateBrepBody Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBrepBody Method (IModeler)

Obsolete. Superseded by [IModeler::ICreateBrepBody3](#).

## .NET Syntax

Visual Basic (Declaration)

```
Function ICreateBrepBody( _  
    ByVal Type As System.Integer, _  
    ByVal NTopologies As System.Integer, _  
    ByRef Topologies As System.Integer, _  
    ByRef EdgeTolArray As System.Double, _  
    ByRef VertexTolArray As System.Double, _  
    ByRef PointArray As System.Double, _  
    ByRef CurveArray As Curve, _  
    ByRef SurfaceArray As Surface, _  
    ByVal NRelations As System.Integer, _  
    ByRef Parents As System.Integer, _  
    ByRef Children As System.Integer, _  
    ByRef Senses As System.Integer _  
) As Body
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Type As System.Integer  
Dim NTopologies As System.Integer  
Dim Topologies As System.Integer  
Dim EdgeTolArray As System.Double  
Dim VertexTolArray As System.Double  
Dim PointArray As System.Double  
Dim CurveArray As Curve  
Dim SurfaceArray As Surface  
Dim NRelations As System.Integer  
Dim Parents As System.Integer  
Dim Children As System.Integer  
Dim Senses As System.Integer  
Dim value As Body
```

```
value = instance.ICreateBrepBody(Type, NTopologies, Topologies, EdgeTolArray, VertexTolArray,  
PointArray, CurveArray, SurfaceArray, NRelations, Parents, Children, Senses)
```

C#

```
Body ICreateBrepBody(  
    System.int Type,  
    System.int NTopologies,  
    ref System.int Topologies,  
    ref System.double EdgeTolArray,  
    ref System.double VertexTolArray,  
    ref System.double PointArray,  
    ref Curve CurveArray,  
    ref Surface SurfaceArray,
```

```
System.int NRelations,  
ref System.int Parents,  
ref System.int Children,  
ref System.int Senses  
)
```

C++/CLI

```
Body^ ICreateBrepBody(  
& System.int Type,  
& System.int NTopologies,  
& System.int% Topologies,  
& System.double% EdgeTolArray,  
& System.double% VertexTolArray,  
& System.double% PointArray,  
& Curve^% CurveArray,  
& Surface^% SurfaceArray,  
& System.int NRelations,  
& System.int% Parents,  
& System.int% Children,  
& System.int% Senses  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Type*

*NTopologies*

*Topologies*

*EdgeTolArray*

*VertexTolArray*

*PointArray*

*CurveArray*

*SurfaceArray*

*NRelations*

*Parents*

*Children*

*Senses*

### Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBrepBody](#).

### See Also

[IModeler Interface](#)

[IModeler Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## ICreateBrepBody2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ICreateBrepBody2 Method (IModeler)

Obsolete. Superseded by [IModeler::ICreateBrepBody3](#).

### .NET Syntax

Visual Basic (Declaration)

```
Function ICreateBrepBody2( _  
    ByVal Type As System.Integer, _  
    ByVal NTopologies As System.Integer, _  
    ByRef Topologies As System.Integer, _  
    ByRef EdgeTolArray As System.Double, _
```

```

ByRef VertexTolArray As System.Double, _
ByRef PointArray As System.Double, _
ByRef CurveArray As Curve, _
ByRef SurfaceArray As Surface, _
ByVal NRelations As System.Integer, _
ByRef Parents As System.Integer, _
ByRef Children As System.Integer, _
ByRef Senses As System.Integer _
) As Body2

```

Visual Basic (Usage)



```

Dim instance As IModeler
Dim Type As System.Integer
Dim NTopologies As System.Integer
Dim Topologies As System.Integer
Dim EdgeTolArray As System.Double
Dim VertexTolArray As System.Double
Dim PointArray As System.Double
Dim CurveArray As Curve
Dim SurfaceArray As Surface
Dim NRelations As System.Integer
Dim Parents As System.Integer
Dim Children As System.Integer
Dim Senses As System.Integer
Dim value As Body2

```

```

value = instance.ICreateBrepBody2(Type, NTopologies, Topologies, EdgeTolArray, VertexTolArray,
PointArray, CurveArray, SurfaceArray, NRelations, Parents, Children, Senses)

```

C#

```

Body2 ICreateBrepBody2(
    System.int Type,
    System.int NTopologies,
    ref System.int Topologies,
    ref System.double EdgeTolArray,
    ref System.double VertexTolArray,
    ref System.double PointArray,
    ref Curve CurveArray,
    ref Surface SurfaceArray,
    System.int NRelations,
    ref System.int Parents,
    ref System.int Children,
    ref System.int Senses
)

```

C++/CLI

```

Body2^ ICreateBrepBody2(
    & System.int Type,
    & System.int NTopologies,
    & System.int% Topologies,
    & System.double% EdgeTolArray,
    & System.double% VertexTolArray,
    & System.double% PointArray,
    & Curve^% CurveArray,
    & Surface^% SurfaceArray,
    & System.int NRelations,
    & System.int% Parents,
    & System.int% Children,
    & System.int% Senses
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Type*

*NTopologies*

*Topologies*

*EdgeToIArray*

*VertexToIArray*

*PointArray*

*CurveArray*

*SurfaceArray*

*NRelations*

*Parents*

*Children*

*Senses*

## Visual Basic for Applications (VBA) Syntax

See [Modeler::ICreateBrepBody2](#).

## See Also

[IModeler Interface](#)

[IModeler Members](#)